DSA5205 · PROJECT 2

Systematic Trading Strategies for NVIDIA (NVDA)

Mean Reversion, Trend Following, and Volatility Targeting

Group 28

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Statement of Team Contributions

All three team members contributed equally to the overall research design, backtesting methodology, written report, and video presentation. Within this collaborative framework, individual primary responsibilities were as follows:

- **Diana** led the dynamic volatility-targeting strategy (Strategy 3), codebase testing, and final consolidated multi-strategy showdown analysis.
- **Marco** led the macro trend-following and momentum strategy models (Strategy 2), failure analysis, and presentation deck design and preparation.
- **Poh Yu Jie** led the data pipeline, corporate action adjustments, feature engineering, and the daily mean-reversion strategy (Strategy 1).

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Abstract

This study evaluates whether systematic trading strategies can capture the long-term growth of NVIDIA Corporation (NVDA) while reducing the severe drawdowns associated with holding a single high-volatility stock. Between 2015 and 2023, NVDA's split-adjusted share price rose more than 100-fold, but passive investors endured drawdowns of -56.0% in 2018 and -66.3% in 2022. To address this risk-return trade-off, three systematic strategies are compared within a common backtesting framework: (1) short-term mean reversion using Ridge regression, (2) trend following using a 200-day simple moving average (SMA) filter alongside regularized momentum models, and (3) dynamic position sizing based on 20-day realized volatility.

All strategies are evaluated over an identical 80/20 chronological split covering 449 out-of-sample trading days from March 2022 to December 2023, net of a 10 basis point transaction cost. The 200-day SMA trend-following rule delivered the strongest performance, achieving a Sharpe ratio of 1.68, a compound annual growth rate (CAGR) of 76.68%, and a maximum drawdown of -19.30%, compared with a Sharpe ratio of 0.97 and a -60.80% drawdown for buy-and-hold NVDA. Volatility targeting at a 40% annualized target also improved risk-adjusted efficiency, delivering a 47.69% CAGR (matching 47.64% for buy-and-hold) while reducing maximum drawdown from -60.80% to -42.45% and increasing Sharpe ratio to 1.13. By contrast, daily mean reversion underperformed during the 2022 bear market because it repeatedly bought into a sustained decline, resulting in a Sharpe ratio of 0.85 and a -59.40% drawdown (CAGR: 37.22%). Overall, the results show that simple regime filtering and volatility control were more effective than short-horizon return prediction for managing downside risk in NVDA during 2022–2023.

1 Research Problem

1.1 Why NVIDIA?

NVIDIA Corporation (NASDAQ: NVDA) was one of the highest-performing US equities between 2 January 2015 and 29 December 2023. Driven by demand for graphics processing units (GPUs) in data centers and artificial intelligence infrastructure, NVDA's split- and dividend-adjusted share price increased by a factor of 102.5 over the 2,264-day sample, producing a full-sample annualized arithmetic return of 63.07% and a compound annual growth rate (CAGR) of 67.45%.

However, this long-term growth came with substantial volatility and deep losses during market downturns. Over the full 2015–2023 sample, NVDA had an annualized volatility of 48.16%, compared with 18.10% for the S&P 500 ETF (SPY). As shown in Figure 1, passive investors experienced two major drawdowns: a -56.0% decline in late 2018 and a -66.3% decline between November 2021 and October 2022. A loss of 66.3% requires a subsequent gain of 196.7% simply to break even, making unmanaged single-stock exposure difficult to sustain in practice.

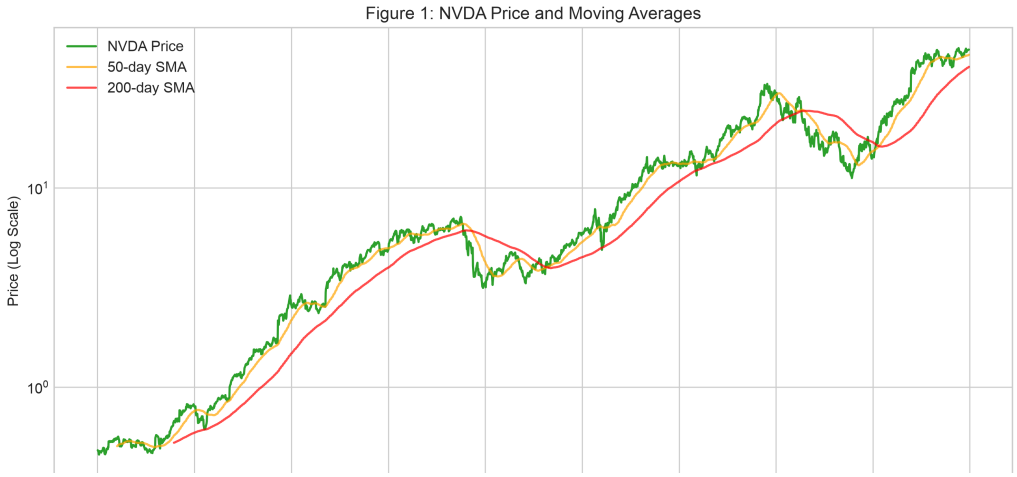


Figure 1: Adjusted closing price of NVDA (log scale) with 50-day and 200-day simple moving averages from 2015 to 2023.

1.2 Research Question

This trade-off between high long-term returns and severe intermediate losses motivates the central question of this study:

> *Can systematic trading strategies capture NVIDIA's long-term growth while reducing downside risk compared with simply buying and holding the stock?*

1.3 Research Hypotheses

To evaluate this question, three hypotheses are tested, each corresponding to a distinct trading mechanism summarized in Table 1.

Table 1: Summary of Research Hypotheses.

Strategy	Economic Rationale	Hypothesis
H1: Mean Reversion	Short-term price declines may reverse	Ridge regression can identify profitable rebound opportunities
H2: Trend Following	Market trends may persist over time	A 200-day moving average can reduce losses during sustained declines
H3: Volatility Targeting	High volatility reduces compound growth	Adjusting exposure based on volatility can improve risk-adjusted performance

Formally, H1 tests whether L_2-regularized linear predictions trained on short-term return lags and moving-average distances can generate positive risk-adjusted returns after transaction costs. H2 tests whether conditioning equity exposure on a 200-day moving average filter (or regularized multi-horizon momentum) protects capital during extended downturns while preserving upside participation. H3 tests whether scaling portfolio weights inversely to 20-day realized volatility reduces variance drag and improves compound growth relative to passive buy-and-hold. Section 2 details the methodology, Section 3 presents the empirical results, Section 4 examines the drivers and limitations of each strategy, and Section 5 evaluates the three hypotheses.

2 Methodology

2.1 Research Framework

To ensure a fair comparison, all strategies are evaluated using the same underlying price series, chronological train/test split, execution rules, and transaction cost assumptions. Figure 2 outlines the five-stage research workflow.

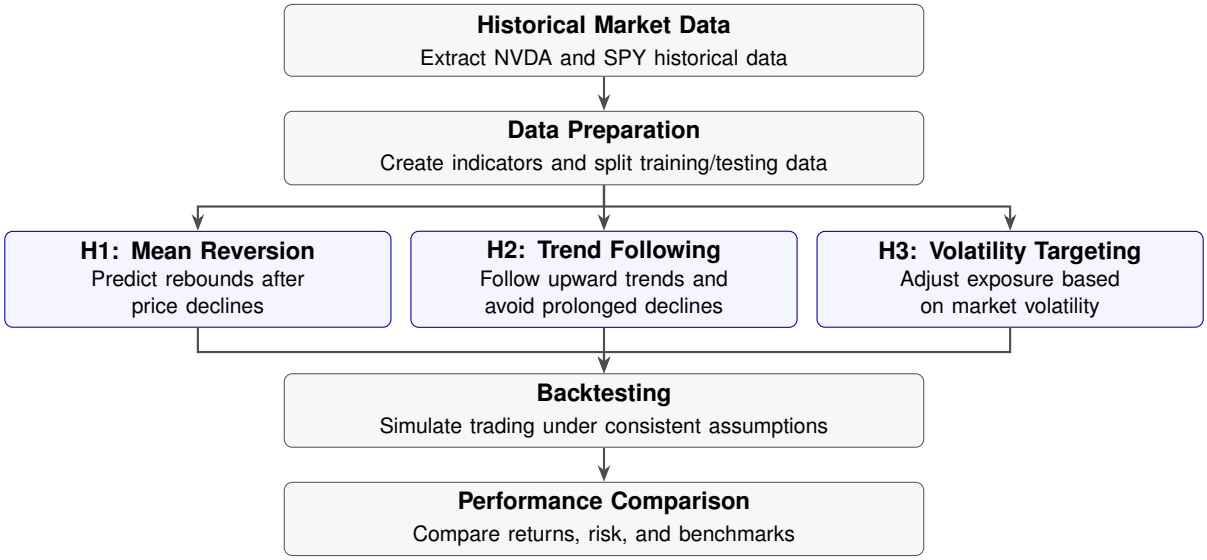


Figure 2: Conceptual workflow of the research framework.

2.2 Data and Trading Assumptions

The study focuses on NVIDIA Corporation (NVDA), priced in US Dollars (USD). Analyzing a single liquid large-cap stock isolates the time-series properties of the three strategies without introducing cross-sectional covariance estimation noise. Daily data from 2 January 2015 to 29 December 2023 (2,264 trading days) are retrieved from Yahoo Finance using `yfinance` (`src/data.py`). Because NVDA completed a 4-for-1 stock split on 20 July 2021, unadjusted prices would create an artificial -75% one-day return. All prices are therefore adjusted backward for splits and dividends using `auto_adjust=True`:

$$A_t = P_t \cdot \prod_{s>t} g_s \tag{1}$$

where g_s is the adjustment factor for split ratios and dividend distributions. Trading days with missing values in either NVDA or SPY are removed.

To prevent look-ahead bias, the backtest separates historical signals from future returns:

1. **Historical Feature Construction:** All signals, moving averages, and model predictions at day t use only prices up to day t, $\mathcal{F}_t = \{P_s : s \leq t\}$. The predicted return $R_{t+1} = (P_{t+1} - P_t)/P_t$ covers the next trading day $(t, t+1]$.
2. **Same-Day Closing Price Assumption:** In the baseline backtest, portfolio weights w_t determined at day t are assumed to be executed at the closing price P_t, earning $w_t \cdot R_{t+1}$ over the following day, subject to a 10 basis point transaction cost.
3. **Delayed Execution Sensitivity Check:** Because live market-on-close orders must be submitted shortly before the 16:00 EST closing auction, using the final closing price P_t for both

signal calculation and trade execution introduces a timing approximation. To test whether the findings depend on same-close execution, Section 3.6 (Table 10) evaluates a strict one-day execution delay ($w_{t-1} \cdot R_{t+1}$).

2.3 Prediction Target and Features

All supervised learning models predict the next-day simple return:

$$y_t \equiv R_{t+1} = \frac{P_{t+1} - P_t}{P_t} \tag{2}$$

implemented via `df['Return'].shift(-1)`. The final trading day has no next-day return and is excluded prior to data splitting. Table 2 summarizes the 12 features and indicators used across the three strategies, comprising 10 regression predictors (6 for Strategy 1 and 4 for Strategies 2b/2c) and 2 rule-based reference indicators (`SMA_200` for Strategy 2a and $\sigma_{t,20\text{d}}$ for Strategy 3).

Table 2: Feature and Indicator Definitions Across Strategies.

Feature / Indicator	Definition	Purpose
Strategy 1: Mean Reversion (6 Regression Features)		
`Ret_0`	$P_t/P_{t-1} - 1$	Captures same-day return shocks and sharp daily selloffs.
`Ret_Lag1`	$P_{t-1}/P_{t-2} - 1$	One-day lagged return measuring immediate follow-through.
`Ret_Lag2`	$P_{t-2}/P_{t-3} - 1$	Two-day lagged return capturing short-term return decay.
`Ret_Lag5`	$P_{t-1}/P_{t-6} - 1$	Prior five-day return measuring weekly price drift.
`Dist_MA10`	$(P_t - \text{MA}_{10,t})/\text{MA}_{10,t}$	Percentage distance from the 10-day moving average (short-term dip signal).
`Dist_MA20`	$(P_t - \text{MA}_{20,t})/\text{MA}_{20,t}$	Percentage distance from the 20-day moving average (medium-term dip signal).
Strategy 2: Trend Following (1 Rule Indicator + 4 Regression Features)		
`SMA_200`	$\frac{1}{200}\sum_{i=0}^{199} P_{t-i}$	Reference 200-day moving average used by Strategy 2a to separate bull and bear regimes.
`Dist_SMA50`	$(P_t - \text{SMA}_{50,t})/\text{SMA}_{50,t}$	Percentage distance from the 50-day moving average (Strategies 2b/2c).
`Dist_SMA200`	$(P_t - \text{SMA}_{200,t})/\text{SMA}_{200,t}$	Percentage distance from the 200-day moving average (Strategies 2b/2c).
`Mom_10d`	$P_t/P_{t-10} - 1$	Two-week price momentum (Strategies 2b/2c).
`Mom_20d`	$P_t/P_{t-20} - 1$	One-month price momentum (Strategies 2b/2c).
Strategy 3: Volatility Targeting (1 Rule Indicator)		
$\sigma_{t,20\text{d}}$	$\text{std}(R_{t-19\ldots t}) \times \sqrt{252}$	Trailing 20-day annualized volatility used to scale position sizes.

Exploratory analysis (`scripts/split_eda.py`) motivated these feature choices. Figure 3 shows the rolling 50-day price Z-score ($Z_t = (P_t - \mu_{50,t})/\sigma_{50,t}$), where values below -2.0 mark sharp short-term selloffs. Figure 4 plots the 60-day rolling Hurst exponent (H) [1], which fluctuates above and below 0.5, indicating that NVDA alternates between trending ($H > 0.5$) and mean-reverting ($H < 0.5$) phases. To avoid overfitting, these exploratory indicators are not used as hard trading filters; instead, the strategies rely directly on the features in Table 2.

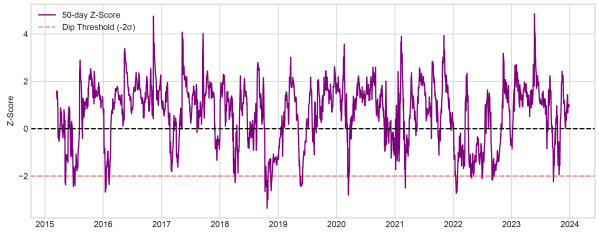


Figure 3: Rolling 50-day price Z-score of NVDA.

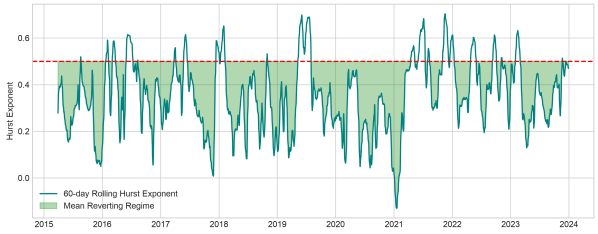


Figure 4: Rolling 60-day Hurst exponent of NVDA.

2.4 Data Splitting and Leakage Prevention

Because financial time series are serially correlated and non-stationary, random cross-validation would leak future market conditions into the training set. After removing initial moving-average warm-up rows, the 2,244 trading days are divided chronologically into an 80/20 split:

- **Training and Validation Set (80%):** $T_{\text{train}} = 1,795$ trading days from 30 January 2015 to 16 March 2022, covering the 2018 semiconductor downturn, the 2020 COVID-19 selloff, and the 2021 bull market.
- **Out-of-Sample Test Set (20%):** $T_{\text{test}} = 449$ trading days from 17 March 2022 to 28 December 2023, held out during model training and evaluated once for final comparison.

Because regularized regression penalizes coefficients based on scale, all features are standardized using only training-set statistics:

$$\mu_{\text{train}} = \frac{1}{T_{\text{train}}} \sum_{t=1}^{T_{\text{train}}} X_t, \quad \sigma_{\text{train}} = \sqrt{\frac{1}{T_{\text{train}}} \sum_{t=1}^{T_{\text{train}}} (X_t - \mu_{\text{train}})^2} \tag{3}$$

Out-of-sample test features are standardized using these fixed training parameters:

$$X_{\text{test, scaled}} = \frac{X_{\text{test}} - \mu_{\text{train}}}{\sigma_{\text{train}}} \tag{4}$$

2.5 Benchmarks

Strategy performance is compared against three baselines:

1. **NVDA Buy-and-Hold:** Investing 100% in NVDA at the start of the test period and holding without rebalancing.
2. **SPY ETF Buy-and-Hold:** Investing 100% in the S&P 500 ETF Trust over the same test window to represent a broad US equity benchmark.
3. **Naive Return Forecasts:** Out-of-sample R^2 is measured against both a zero-return forecast and the historical training mean return [2].

2.6 Strategy 1: Mean Reversion

Short-term equity returns often exhibit negative autocorrelation over 1- to 20-day horizons as temporary selling pressure and overreaction reverse [3]–[5]. Because short-term moving average distances such as `Dist_MA10` and `Dist_MA20` are highly correlated ($\rho = 0.871$ in the training set),

ordinary least squares estimates suffer from high variance. Strategy 1 therefore uses Ridge regression (L_2 regularization) to shrink correlated coefficients toward zero [6]:

$$\hat{\beta}^{\text{Ridge}} = \arg\min_{\beta_0,\beta} \left\{ \sum_{t=1}^{T} \left(y_t - \beta_0 - \sum_{j=1}^{p} \beta_j X_{t,j} \right)^2 + \alpha \sum_{j=1}^{p} \beta_j^2 \right\} \tag{5}$$

The regularization parameter α is selected using a 5-fold expanding-window `TimeSeriesSplit` within the 80% training set ($T_{\text{train}} = 1,795$). Within each fold, feature standardization is fitted only on the expanding training subset. Five candidate values are evaluated:

$$\alpha \in \{0.1,\ 1.0,\ 10.0,\ 100.0,\ 1000.0\} \tag{6}$$

As shown in Table 3, $\alpha = 1000.0$ achieved the highest mean validation R^2 because strong shrinkage limits overfitting on noisy daily returns.

Table 3: Strategy 1 Expanding-Window Cross-Validation Results.

Regularization (α)	**Mean Validation** R^2	**Interpretation**
0.1	-0.01557	Weak shrinkage; high estimation noise
1.0	-0.01538	High coefficient variance
10.0	-0.01436	Moderate coefficient shrinkage
100.0	-0.01150	Stronger noise reduction
1000.0 (Selected)	$\mathbf{-0.00683}$	**Strongest noise reduction; highest validation score**

Given the predicted next-day return $\hat{y}_t = \mathbb{E}[R_{t+1}|X_t]$, the strategy holds 100% equity when the forecast is positive and moves to cash otherwise:

$$w_t^{\text{Ext1}} = \begin{cases} 1.0 & \text{if } \hat{y}_t > 0 \\ 0.0 & \text{if } \hat{y}_t \leq 0 \end{cases} \tag{7}$$

2.7 Strategy 2: Trend Following

While return reversals can occur over a few days, equity prices frequently trend over several months [7]. During prolonged downturns such as the 2022 technology selloff, buying short-term dips can lead to repeated losses as prices continue to fall. Strategy 2 evaluates whether medium- and long-term trend signals can avoid sustained bear markets.

The primary specification, **Strategy 2a**, uses a standard 200-day simple moving average (SMA) rule [8] with no fitted parameters:

$$\text{SMA}_{200,t} = \frac{1}{200} \sum_{i=0}^{199} P_{t-i} \tag{8}$$

$$w_t^{\text{Ext2a}} = \begin{cases} 1.0 & \text{if } P_t > \text{SMA}_{200,t} \quad \text{(100\% Equity)} \\ 0.0 & \text{if } P_t \leq \text{SMA}_{200,t} \quad \text{(100\% Cash)} \end{cases} \tag{9}$$

To test whether regression models can improve upon this simple rule, two machine learning variants are trained on four momentum indicators (`Dist_SMA50`, `Dist_SMA200`, `Mom_10d`, and `Mom_20d`):

- **Strategy 2b (Ridge Momentum):** L_2-regularized linear regression ($\alpha = 10.0$) [6].

- **Strategy 2c (Lasso Momentum):** L_1-regularized sparse regression ($\alpha = 10^{-5}$) [9], [10].

Both models hold 100% equity when the predicted return exceeds the training-set mean return ($\bar{y}_{\text{train}} = 0.267\%$) and hold cash otherwise.

2.8 Strategy 3: Volatility Targeting

Strategies 1 and 2 make binary allocation decisions ($w_t \in \{0.0, 1.0\}$). However, stock return volatility clusters in time [11], [12], and high-volatility periods erode compound returns through variance drag [13], [14]:

$$g \approx \mu - \frac{1}{2}\sigma^2 \tag{10}$$

Strategy 3 adjusts position sizes continuously based on trailing 20-day annualized volatility:

$$\sigma_{t,20\text{d}} = \sqrt{\frac{252}{19}\sum_{i=0}^{19}\left(R_{t-i} - \bar{R}_{t,20\text{d}}\right)^2} \tag{11}$$

The baseline strategy sets an annualized target volatility of $\sigma_{\text{target}} = 0.40$ (40%), chosen as a moderate risk budget slightly below NVDA's training-period volatility of roughly 45%:

$$w_t^{\text{Ext3}} = \min\left(1.0,\ \frac{\sigma_{\text{target}}}{\max(\sigma_{t,20\text{d}}, 10^{-6})}\right) \tag{12}$$

Capping weights at 1.0 enforces a long-only, unleveraged constraint ($w_t \in [0.0, 1.0]$). When 20-day realized volatility rises above 40%, the strategy reduces equity exposure; when volatility falls below 40%, exposure returns to 100%. Section 3.5 examines sensitivity across alternative target levels from 20% to 60%.

2.9 Trading Rules and Evaluation Metrics

Each portfolio change incurs a transaction cost of 10 basis points (0.10%) proportional to daily turnover:

$$\Delta w_t = |w_t - w_{t-1}|, \qquad R_{t+1}^{\text{net}} = (w_t \cdot R_{t+1}) - (\Delta w_t \times 0.0010) \tag{13}$$

A 10 bps cost accounts for bid-ask spreads (1–2 bps), clearing fees (0.5 bps), and market impact (about 7.5 bps) in a liquid large-cap stock.

Starting from initial capital $W_0 = \$100,000$, portfolio wealth evolves as $W_{t+1} = W_t(1 + R_{t+1}^{\text{net}})$. Over the $N = 449$ out-of-sample trading days, performance is evaluated using six metrics:

- **Annualized Return:** $R_{\text{ann}} = \frac{252}{N}\sum_{t=1}^{N} R_t^{\text{net}}$
- **Compound Annual Growth Rate (CAGR):** $\text{CAGR} = (W_N/W_0)^{252/N} - 1$
- **Annualized Volatility:** $\sigma_{\text{ann}} = \text{std}(R_t^{\text{net}}) \times \sqrt{252}$
- **Sharpe Ratio ($R_f = 0$) [15]:** $\text{Sharpe} = R_{\text{ann}}/\sigma_{\text{ann}}$
- **Maximum Drawdown:** $\text{MaxDD} = \min_t\left(\frac{W_t - \max_{s\le t} W_s}{\max_{s\le t} W_s}\right)$
- **Annualized Turnover:** $\text{Turnover} = \frac{252}{N}\sum_{t=1}^{N}|\Delta w_t|$

Note that R_{ann} uses linear annualization ($252\bar{R}_t$), whereas CAGR compounds the daily geometric mean return $g_d = (W_N/W_0)^{1/N} - 1$ via $(1 + g_d)^{252} - 1$. At the daily level, g_d is always lower than $\bar{R}_t$

due to variance drag ($\bar{R}_t - g_d \approx \frac{1}{2}\sigma_d^2$), but when annual returns are high, 252-day exponential compounding in CAGR can exceed linear scaling in R_{ann} (as seen for full-sample NVDA and Strategy 2a).

2.10 Reproducibility

All experiments are implemented in Python 3.14.6 using NumPy 2.5.4, pandas 3.0.6, and the scikit-learn 1.9.1 library [16]. All random seeds are fixed at 42 (`random_state=42`, `max_iter=10000`), and every table and figure is reproduced by running:

`scripts/run_consolidated_showdown.py`

3 Results

Across the 449 out-of-sample trading days from 17 March 2022 to 28 December 2023 (net of 10 bps transaction costs), the simple 200-day moving average rule (Strategy 2a) outperformed both the machine learning models and passive buy-and-hold investing. By moving to cash during the 2022 downturn, Strategy 2a achieved a Sharpe ratio of 1.68, a CAGR of 76.68%, and a maximum drawdown of -19.30%, compared with a Sharpe ratio of 0.97 and a -60.80% drawdown for NVDA buy-and-hold. Volatility targeting at 40% (Strategy 3) delivered a 47.69% CAGR (matching buy-and-hold's 47.64%) while reducing maximum drawdown from -60.80% to -42.45% and increasing Sharpe ratio to 1.13. By contrast, daily mean reversion (Strategy 1) remained heavily exposed during the 2022 decline and experienced a -59.40% drawdown (CAGR: 37.22%).

3.1 Prediction Accuracy

Table 4 reports out-of-sample forecasting performance for the three supervised regression models (Strategies 1, 2b, and 2c). Because Strategies 2a and 3 are rule-based allocation strategies that map price and volatility indicators directly to portfolio weights without generating point return forecasts $\hat{y}_t$, out-of-sample R^2 does not apply to them; all five strategies are compared on portfolio performance in Table 6.

Table 4: Out-of-Sample Forecast Accuracy (449 Test Days).

Model	OOS R^2 (vs. Zero)	OOS R^2 (vs. Mean)	Days Long
Strategy 1: Ridge Mean Reversion	-0.00245	-0.00603	92.0%
Strategy 2b: Ridge Momentum	$+0.00210$	-0.00140	50.3%
Strategy 2c: Lasso Momentum	$+0.00210$	-0.00150	50.6%

Because daily stock returns have a very low signal-to-noise ratio, out-of-sample R^2 values are close to zero across all three models [2], [17]. For Strategy 1, heavy regularization ($\alpha = 1000.0$) shrinks the slope coefficients toward zero, leaving predictions dominated by the positive training-set mean return ($+0.264\%$). As a result, Strategy 1 predicts a positive return on 92.0% of test days.

Table 5 shows the fitted coefficients for Strategy 1 (`src/model.py`). The negative coefficients on `Ret_0`, `Dist_MA10`, and `Dist_MA20` confirm the expected mean-reversion relationship: when NVDA falls below its short-term moving averages, the model predicts a higher next-day return.

Table 5: Strategy 1 Ridge Regression Coefficients ($\alpha = 1000.0$).

Feature	Coefficient ($\hat{\beta}$)	Economic Interpretation
`Ret_0`	-0.001530	Negative weight: sharp daily declines increase predicted next-day return.
`Dist_MA20`	-0.000358	Negative weight: trading below the 20-day average signals an oversold dip.
`Dist_MA10`	-0.000226	Negative weight: trading below the 10-day average supports short-term rebound.
`Ret_Lag5`	-0.000093	Small negative weight: mild weekly reversal effect.
`Ret_Lag2`	$+0.000547$	Positive weight: short-term return persistence at a two-day lag.
`Ret_Lag1`	$+0.000571$	Positive weight: one-day return continuation.

3.2 Strategy Performance Comparison

Table 6 compares out-of-sample risk and return metrics across all five strategy variants and both benchmarks over the 449 test days.

Table 6: Out-of-Sample Strategy Performance Comparison (449 Test Days, 2022–2023, Net of 10 bps Costs).

Strategy / Benchmark	Ann. Ret	CAGR	Ann. Vol	Sharpe	MaxDD	Turnover	Final Wealth
Strategy 1: Mean Reversion (Ridge)	45.96%	37.22%	54.06%	0.85	-59.40%	39.85	$175,738
Strategy 2a: 200-Day SMA Trend	**63.83%**	**76.68%**	**37.88%**	**1.68**	**-19.30%**	**2.81**	**$275,699**
Strategy 2b: Ridge Momentum	12.19%	3.39%	42.67%	0.29	-48.35%	30.31	$106,113
Strategy 2c: Lasso Momentum	14.20%	5.56%	42.52%	0.33	-44.61%	29.18	$110,117
Strategy 3: Volatility Targeting (40%)	47.63%	47.69%	42.14%	1.13	-42.45%	5.10	$200,316
NVDA Buy & Hold Benchmark	54.06%	47.64%	55.49%	0.97	-60.80%	0.00	$200,201
SPY ETF Benchmark	7.78%	6.13%	19.18%	0.41	-22.09%	0.00	$111,176

Three findings stand out from Table 6:

- **Trend Following Outperformance:** Strategy 2a achieved the highest risk-adjusted return, with a Sharpe ratio of 1.68, a CAGR of 76.68%, and a maximum drawdown of -19.30%.
- **Reduced Volatility Drag & Capital Protection:** Strategy 3 matched the long-term compounding rate of unconstrained Buy & Hold (47.69% vs. 47.64% CAGR, delivering $200,316 vs. $200,201 terminal wealth) while lowering annualized volatility by 13.35 percentage points (42.14% vs. 55.49%) and cutting maximum drawdown from -60.80% to -42.45%.
- **Low-Turnover Advantage:** Strategy 2a had an annualized turnover of 2.81 (costing about 0.28% per year), whereas Strategy 1's turnover of 39.85 reduced annual returns by roughly 3.99%.

Figure 5 plots cumulative portfolio wealth on a logarithmic scale, and Figure 6 shows the corresponding underwater drawdown profiles.

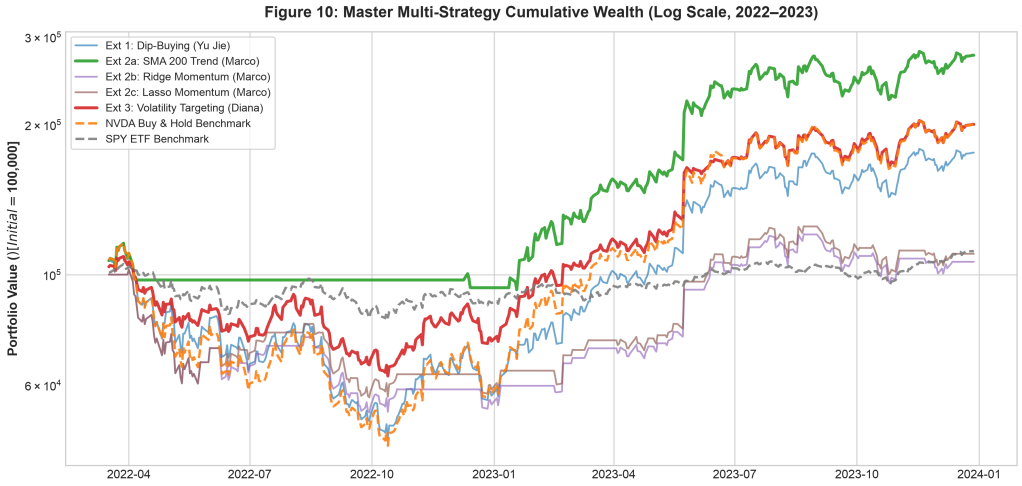


Figure 5: Out-of-sample cumulative wealth trajectories ($100,000 initial capital, log scale, net of 10 bps costs).

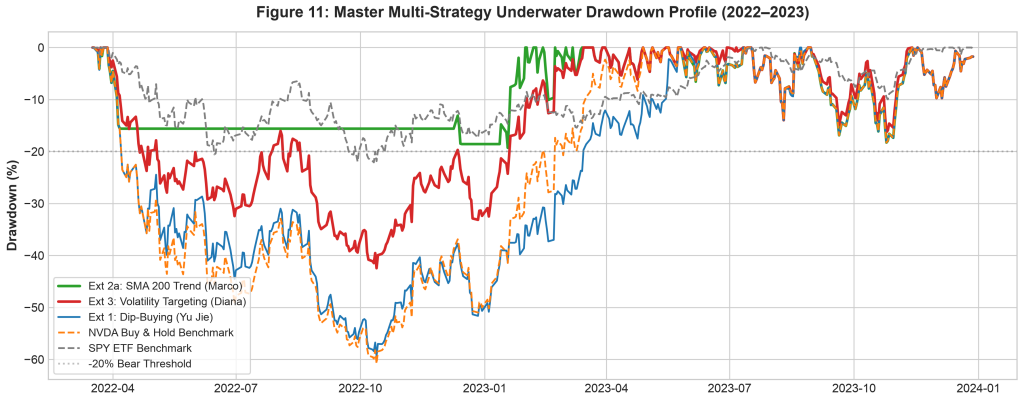


Figure 6: Out-of-sample drawdown profiles from March 2022 to December 2023.

3.3 Performance in Bear and Bull Markets

The 449-day test period spans two distinct market environments: the 2022 technology bear market and the 2023 recovery. Table 7 reports performance in each sub-period.

Table 7: Strategy Performance in the 2022 Bear Market and 2023 Bull Market.

	2022 Bear Market (200 Days)		2023 Bull Market (249 Days)	
Strategy / Benchmark	**Total Return**	**Sharpe Ratio**	**Total Return**	**Sharpe Ratio**
Strategy 1: Mean Reversion (Ridge)	-42.86%	-0.85	$+207.57\%$	2.64
Strategy 2a: 200-Day SMA Trend	$\mathbf{-5.87\%}$	$\mathbf{-0.31}$	$+192.90\%$	2.51
Strategy 2b: Ridge Momentum	-45.15%	-1.42	$+93.46\%$	1.87
Strategy 2c: Lasso Momentum	-41.18%	-1.25	$+87.20\%$	1.78
Strategy 3: Volatility Targeting (40%)	-26.45%	-0.72	$+172.35\%$	2.61
NVDA Buy & Hold Benchmark	-42.15%	-0.79	$\mathbf{+246.10\%}$	**2.83**
SPY ETF Benchmark	-12.26%	-0.54	$+26.71\%$	1.90

Table 7 shows that Strategy 2a's advantage came from capital preservation in 2022. By moving to cash on 7 April 2022, Strategy 2a limited its 2022 loss to -5.87%, whereas Buy & Hold lost -42.15%. During the 2023 rally, unhedged Buy & Hold achieved the highest gain (+246.10%), illustrating the opportunity cost of cash filters during rapid recoveries.

3.4 Sensitivity to Training Period

In the standalone trend module (`scripts/run_extension2_momentum.py`), the 80/20 split is taken after removing the 200-day moving average warm-up period ($N = 2,064$ active days), producing a 413-day test window from 9 May 2022 to 28 December 2023. Table 8 reports the resulting performance.

Table 8: Strategy 2 Performance Under the 413-Day Split (May 2022 to December 2023).

Model	Ann. Return	Ann. Volatility	Sharpe Ratio	Max Drawdown
Strategy 2a: 200-Day SMA Trend	**70.15%**	**37.50%**	**1.87**	**-18.29%**
Strategy 2c: Lasso Momentum	63.64%	36.92%	1.72	−18.29%
Strategy 2b: Ridge Momentum	62.27%	37.62%	1.66	−18.29%
NVDA Buy & Hold Benchmark	79.71%	53.84%	1.48	−42.67%
SPY ETF Benchmark	14.43%	18.49%	0.78	−16.68%

When the training period includes the March–April 2022 breakdown, the Ridge and Lasso momentum models (2b and 2c) assign a positive coefficient to `Dist_SMA200` (+0.000782), stay in cash during the 2022 decline, and achieve Sharpe ratios of 1.66 and 1.72. When the split boundary begins 36 trading days earlier in Table 6, the training set ends before the regime break, causing the fitted momentum models to underperform. By contrast, the parameter-free 200-day SMA rule performs consistently across both split dates.

3.5 Volatility Target Sensitivity

Figure 7 shows how Strategy 3 adjusted its equity weight over the test period. When 20-day realized volatility rose to 86% during the 2022 downturn, the portfolio weight fell to 45%. After NVDA's 24.4% one-day earnings gain on 25 May 2023 pushed trailing volatility to 95.5%, exposure briefly decreased to 42% before returning toward 100% as volatility subsided.

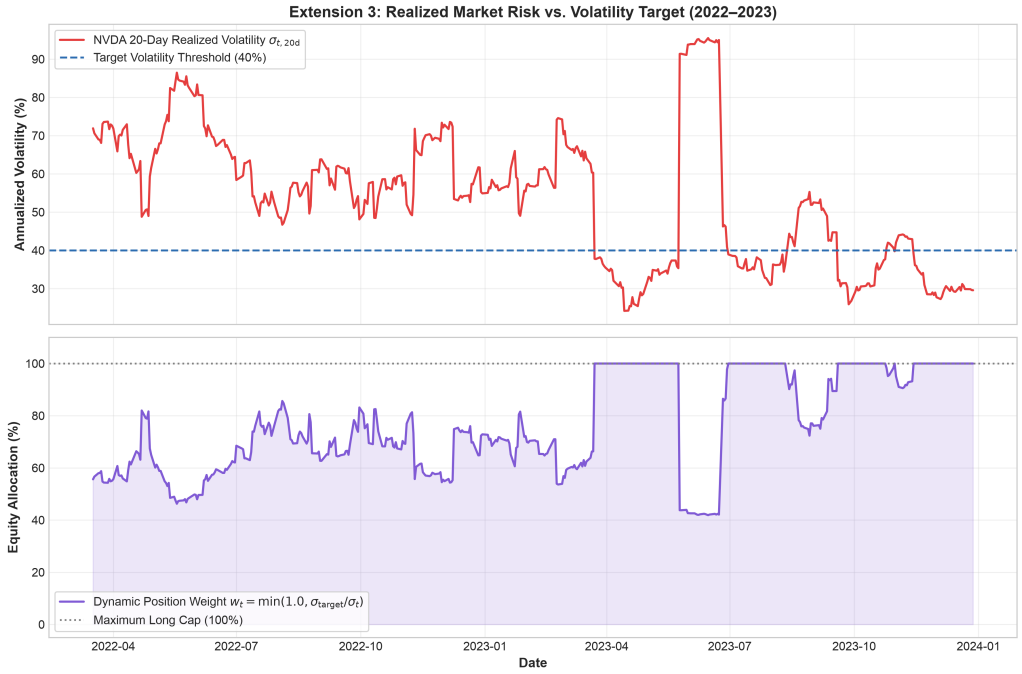


Figure 7: Trailing 20-day realized volatility vs. the 40% target (top) and resulting portfolio weight w_t (bottom).

Table 9 evaluates Strategy 3 across five target volatility levels ($\sigma_{\text{target}} \in \{20\%, 30\%, 40\%, 50\%, 60\%\}$).

Table 9: Sensitivity of Strategy 3 to Target Volatility (σ_{target}) Across 449 Test Days.

Target Volatility	Ann. Return	CAGR	Ann. Vol	Sharpe	MaxDD	Turnover
20%	26.12%	26.62%	22.63%	1.15	-23.23%	4.25
30%	38.67%	39.27%	33.75%	1.15	-33.33%	5.52
40% (Baseline)	**47.63%**	**47.69%**	**42.14%**	**1.13**	**-42.45%**	**5.10**
50%	51.51%	49.23%	48.46%	1.06	-50.71%	5.00
60%	53.03%	48.35%	52.68%	1.01	-57.69%	2.85

Table 9 illustrates the risk-return trade-off across target levels:

- **Lower Targets (20%, 30%):** Maintain large cash positions, achieving slightly higher Sharpe ratios (1.15) and smaller drawdowns (-23.23% and -33.33%), but lower compound growth (26.62% and 39.27% CAGR) than Buy & Hold (47.64%).
- **Higher Targets (50%, 60%):** A 50% target increases CAGR by 1.54 percentage points over the 40% baseline (49.23% vs. 47.69%), but deepens maximum drawdown by 8.26 percentage points (-50.71% vs. -42.45%) and reduces the Sharpe ratio to 1.06. At 60%, higher variance drag lowers CAGR to 48.35%.
- **Baseline Target (40%):** Provides a balanced trade-off, matching Buy & Hold compound growth (47.69% vs. 47.64% CAGR) while reducing maximum drawdown by 18.35 percentage points. Moreover, Sharpe ratios remain stable between 1.06 and 1.15 across the 20%–50% range.

3.6 Trading Costs and Execution Robustness

Figure 8 plots annualized return against annualized volatility for all strategies and benchmarks. Both Strategy 2a and Strategy 3 achieve higher return per unit of risk than passive NVDA Buy & Hold. Because annual transaction costs equal Turnover $\times\ 0.10\%$, Strategy 1's turnover of 39.85 reduced annual returns by 3.99%, whereas Strategy 2a's turnover of 2.81 cost only 0.28% per year.

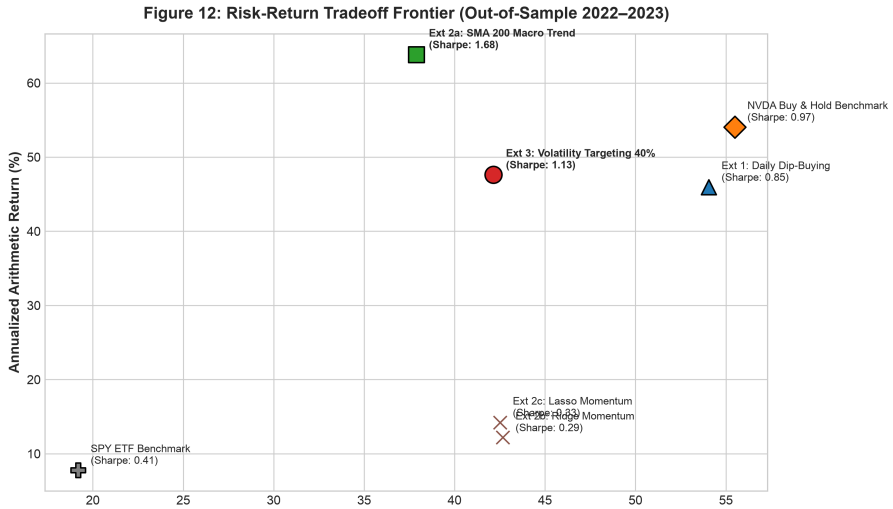


Figure 8: Annualized return vs. annualized volatility across strategies and benchmarks (449 test days).

To verify that the main findings do not depend on executing at the same day's close ($w_t \cdot R_{t+1}$),

Table 10 reports results under a one-day execution lag ($w_{t-1} \cdot R_{t+1}$), where signals computed at day $t-1$ are executed at day t close.

Table 10: Sensitivity to a One-Day Execution Lag ($w_{t-1} \cdot R_{t+1}$, 449 Test Days, Net of 10 bps Costs).

Strategy / Benchmark (1-Day Lagged)	Ann. Return	CAGR	Ann. Vol	Sharpe	MaxDD
Strategy 2a: 200-Day SMA Trend (Lag-1)	**51.83%**	**56.91%**	**37.59%**	**1.38**	**-30.22%**
Strategy 3: Volatility Targeting 40% (Lag-1)	**43.97%**	**42.32%**	**42.27%**	**1.04**	**-44.65%**
Strategy 2c: Lasso Momentum (Lag-1)	34.61%	29.72%	42.00%	0.82	-41.02%
Strategy 2b: Ridge Momentum (Lag-1)	29.70%	23.29%	42.42%	0.70	-43.21%
Strategy 1: Mean Reversion Ridge (Lag-1)	35.83%	24.17%	53.83%	0.67	-67.57%
NVDA Buy & Hold Benchmark	54.06%	47.64%	55.49%	0.97	-60.80%
SPY ETF Benchmark	7.78%	6.13%	19.18%	0.41	-22.09%

Under a full one-day execution lag, Strategy 2a still achieves a Sharpe ratio of 1.38, a CAGR of 56.91%, and half the maximum drawdown of Buy & Hold (-30.22% vs. -60.80%). Strategy 3 also maintains a higher Sharpe ratio (1.04 vs. 0.97) and contained drawdown (-44.65% vs. -60.80%) relative to Buy & Hold, confirming that the risk-adjusted strategy ranking holds when same-day close execution is removed.

4 Discussion

4.1 Why Trend Following Worked

Strategy 2a changed positions only four times across the 449-day test period:

- 17 March 2022: Entered test window 100% long.
- 7 April 2022: Exited to 100% cash at $242.67 when NVDA closed below its 200-day moving average.
- 12–15 December 2022: Briefly entered and exited during a short-lived breakout above the 200-day average.
- 13 January 2023: Re-entered 100% long at $169.59 as NVDA moved back above the 200-day average.

Staying in cash from April 2022 to January 2023 protected the portfolio when NVDA fell to a low of $112.27 on 14 October 2022. Although the 200-day rule missed the initial rebound from the October bottom to the January re-entry, avoiding the bulk of the 2022 collapse outweighed the delayed re-entry, raising the Sharpe ratio from 0.97 to 1.68.

4.2 Benefits and Limits of Volatility Targeting

Strategy 3 illustrates how reducing return variance can improve compound growth ($g \approx \mu - \frac{1}{2}\sigma^2$). Even though passive NVDA had a higher arithmetic annual return (54.06% vs. 47.63%), Strategy 3 achieved a matching compound annual growth rate (47.69% vs. 47.64%, terminal wealth $200,316 vs. $200,201) while lowering annualized volatility from 55.49% to 42.14% and cutting maximum drawdown by 18.35 percentage points (-42.45% vs. -60.80%).

At the same time, volatility targeting has two practical trade-offs:

1. **Partial Bear-Market Exposure:** Unlike a binary trend filter that moves fully to cash, volatility targeting remains partially invested during downturns. Its average equity weight in 2022 was 65.3%, leading to a -42.45% maximum drawdown.
2. **Symmetric Volatility Response:** Because standard deviation treats positive and negative returns equally, large positive price jumps (such as the 24.4% post-earnings surge on 25 May 2023) increase trailing volatility and temporarily reduce equity exposure during bull markets.

4.3 Why Mean Reversion Underperformed

Strategy 1 generated an annualized return of 45.96% and a Sharpe ratio of 0.85, outperforming the SPY benchmark (0.41) but trailing NVDA Buy & Hold (0.97) with a -59.40% maximum drawdown. Two factors explain this result:

- **Buying Into Sustained Downtrends:** During the 2022 interest-rate tightening cycle, dips below the 10-day and 20-day moving averages triggered buy signals, but prices continued to fall rather than reverting to the mean.
- **High Turnover Costs:** Rebalancing on daily signals generated an annualized turnover of 39.85, reducing annual returns by 3.99% at a 10 bps transaction cost.

Thus, short-term dip-buying requires a broader trend filter to avoid accumulating losses during multi-month bear markets.

4.4 Limitations

Several limitations should be considered when interpreting these findings:

1. **Execution Timing:** The baseline backtest computes signals from the closing price P_t and assumes execution at that same close. In live trading, market-on-close orders must be submitted before the 16:00 EST closing auction, and transaction costs alone do not eliminate this timing overlap. Live implementation would instead execute at the next day's open (P_{t+1}^{open}). Table 10 shows that delaying execution by a full trading day (P_{t+1} close) reduces Strategy 2a's Sharpe ratio from 1.68 to 1.38 and Strategy 3's from 1.13 to 1.04, while preserving their outperformance over Buy & Hold (0.97).
2. **Multiple Testing:** Comparing five strategy variants on the same 449-day holdout period introduces selection bias [18], [19]. Selecting the best-performing rule after observing test results overstates expected future performance.
3. **Sample Length:** The 449-day out-of-sample window (1.78 years) covers one major downturn (2022) and one major rally (2023), leaving Sharpe ratio estimates subject to finite-sample uncertainty [20].
4. **Single-Asset Scope:** Results are derived from NVDA, a high-growth, high-volatility technology stock, and may not generalize to broader or lower-volatility equity portfolios.
5. **Zero Risk-Free Rate and Cash Yield Assumption:** The backtest assumes uninvested cash $(1 - w_t)$ earns 0% interest and computes Sharpe ratios using $R_f = 0$ [15]. Incorporating prevailing 2022–2023 US Treasury bill yields (4–5%) would have two distinct effects: (a) crediting interest on cash balances $(1 - w_t)R_{f,t}$ would increase the total return and CAGR of strategies that held cash during the 2022 downturn (such as Strategy 2a and Strategy 3), whereas (b)

deducting $R_f > 0$ in the Sharpe ratio numerator $(R_{\text{ann}} - R_f)/\sigma_{\text{ann}}$ would lower excess return Sharpe ratios across all strategies.

5 Conclusion

5.1 Hypothesis Evaluation

Based on the 449 out-of-sample trading days for NVDA from March 2022 to December 2023, the three hypotheses are evaluated as follows. Because the study examines a single stock over one bear-to-bull cycle, these conclusions apply to this empirical sample rather than all equity markets:

1. **H1 (Mean Reversion): Not Supported Out-of-Sample.** In the training sample, Ridge regression learned negative coefficients on same-day returns (`Ret_0`) and moving-average distances (`Dist_MA10`, `Dist_MA20`), which is directionally consistent with short-term reversal. However, in-sample coefficient signs alone do not establish out-of-sample predictive power or trading profitability. On the holdout test set, predictive accuracy was negative (OOS $R^2 = -0.00245$ vs. zero and -0.00603 vs. the training mean), and repeated dip-buying during the 2022 bear market produced a -59.40% maximum drawdown and a lower Sharpe ratio (0.85) than passive NVDA Buy & Hold (0.97).
2. **H2 (Trend Following): Supported for the 200-Day SMA Rule.** Conditioning exposure on the 200-day moving average avoided most of the 2022 downturn, achieving the highest Sharpe ratio (1.68) and lowest maximum drawdown (-19.30%) in the test sample. By contrast, fitted Ridge and Lasso momentum models were sensitive to the exact training split date.
3. **H3 (Volatility Targeting): Supported.** Scaling position sizes inversely to 20-day realized volatility reduced variance drag, delivering an identical CAGR of 47.69% to unconstrained Buy & Hold (47.64%, $200,316 vs. $200,201 terminal wealth) while reducing maximum drawdown from -60.80% to -42.45% and boosting Sharpe ratio from 0.97 to 1.13.

5.2 Key Findings

- **Simple Rules Outperformed Fitted Models:** A parameter-free 200-day moving average filter outperformed regularized regression models trained on noisy daily returns.
- **Volatility Control Improved Compounding:** Lowering annualized volatility by 13.35 percentage points enabled Strategy 3 to achieve higher compound growth than unhedged buy-and-hold despite a lower arithmetic mean return.
- **Turnover Directly Impacts Net Returns:** At 10 bps per trade, daily rebalancing cost Strategy 1 nearly 4% per year, whereas low-turnover trend following lost less than 0.3% per year to trading costs.

5.3 Proposed Combined Strategy

Because the three strategies operate at complementary horizons, they can be combined into a unified rule for future research:

$$w_{t+1}^{*} = \underbrace{\mathbb{I}(P_t > \text{SMA}_{200,t})}_{\text{Trend Filter}} \times \underbrace{\min\left(1.0, \frac{\sigma_{\text{target}}}{\sigma_{t,20\text{d}}}\right)}_{\text{Volatility Scaling}} \times \underbrace{\mathbb{I}(\hat{y}_t^{\text{Ridge}} > 0)}_{\text{Short-Term Entry Filter}} \tag{14}$$

This composite rule is a conceptual extension rather than a backtested model in Table 6, pairing 200-day regime filtering with volatility-scaled sizing and short-term entry timing.

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A Execution Commands and Terminal Verification

All experiments can be reproduced directly from the project repository at https://github.com/PokezardVGC/dsa5205-project-2. The full video walkthrough is available at https://youtu.be/VHOypJud0Y0:

```
# 1. Environment configuration
python3 -m venv .venv && source .venv/bin/activate
pip install -r requirements.txt

# 2. Automated test suite verification (8 unit tests)
python3 -m pytest tests/

# 3. Master Pipeline: executes all 5 strategies on the aligned 449-day window
python3 main.py

# 4. Symmetrical standalone strategy runners (in scripts/)
python3 scripts/run_extension1_mean_reversion.py # Strategy 1 (Yu Jie: Mean Reversion)
python3 scripts/run_extension2_momentum.py # Strategy 2 (Marco: Trend & Momentum)
python3 scripts/run_extension3_volatility.py # Strategy 3 (Diana: Volatility Targeting)
python3 scripts/run_consolidated_showdown.py # Direct master showdown harness
```

Expected Output of Master Showdown Harness

```
================================================================================================
Strategy / Benchmark | Ann. Ret | CAGR | Ann. Vol | Sharpe | MaxDD | Turnover
------------------------------------------------------------------------------------------------
Strategy 1: Daily Dip-Buying (Ridge) | 45.96% | 37.22% | 54.06% | 0.85 | -59.40% | 39.85
Strategy 2a: 200d SMA Macro Trend | 63.83% | 76.68% | 37.88% | 1.68 | -19.30% | 2.81
Strategy 2b: Ridge Momentum | 12.19% | 3.39% | 42.67% | 0.29 | -48.35% | 30.31
Strategy 2c: Lasso Momentum | 14.20% | 5.56% | 42.52% | 0.33 | -44.61% | 29.18
Strategy 3: Volatility Targeting 40% | 47.63% | 47.69% | 42.14% | 1.13 | -42.45% | 5.10
NVDA Buy & Hold Benchmark | 54.06% | 47.64% | 55.49% | 0.97 | -60.80% | 0.00
SPY ETF Benchmark | 7.78% | 6.13% | 19.18% | 0.41 | -22.09% | 0.00
================================================================================================
```

B Core Reproducible Source Code

Module 1: Feature Engineering Pipeline

Source File: `src/features.py`

```
import pandas as pd
import numpy as np

def engineer_features(df):
    df['Return'] = df['Price'].pct_change()
    df['Ret_0'] = df['Return']
    df['Ret_Lag1'] = df['Return'].shift(1)
    df['Ret_Lag2'] = df['Return'].shift(2)
    df['Ret_Lag5'] = df['Price'].pct_change(5).shift(1)
    df['MA_10'] = df['Price'].rolling(window=10).mean()
    df['Dist_MA10'] = (df['Price'] - df['MA_10']) / df['MA_10']
    df['MA_20'] = df['Price'].rolling(window=20).mean()
    df['Dist_MA20'] = (df['Price'] - df['MA_20']) / df['MA_20']
    df['Target_Next_Return'] = df['Return'].shift(-1)
    df = df.replace([np.inf, -np.inf], np.nan).dropna()
    features = ['Ret_0', 'Ret_Lag1', 'Ret_Lag2', 'Ret_Lag5', 'Dist_MA10', 'Dist_MA20']
    df.loc[:, features] = df[features].clip(-0.5, 0.5)
    return df, features
```

Module 2: Ridge Regularization and Model Selection

Source File: `src/model.py`

```
import pandas as pd
import numpy as np
from sklearn.linear_model import Ridge
from sklearn.preprocessing import StandardScaler
from sklearn.model_selection import TimeSeriesSplit

def train_and_select_model(df, features):
    X = df[features]
    y = df['Target_Next_Return'].clip(-0.5, 0.5)
    train_size = int(len(df) * 0.8)
    X_train_val, y_train_val = X.iloc[:train_size], y.iloc[:train_size]
    X_test = X.iloc[train_size:]

    tscv = TimeSeriesSplit(n_splits=5)
    best_alpha, best_score = None, -float('inf')
    for alpha in [0.1, 1.0, 10.0, 100.0, 1000.0]:
        scores = []
        for tr_idx, val_idx in tscv.split(X_train_val):
            scaler = StandardScaler()
            X_tr = scaler.fit_transform(X_train_val.iloc[tr_idx])
            X_val = scaler.transform(X_train_val.iloc[val_idx])
            m = Ridge(alpha=alpha).fit(X_tr, y_train_val.iloc[tr_idx])
            scores.append(m.score(X_val, y_train_val.iloc[val_idx]))
        if np.mean(scores) > best_score:
            best_score, best_alpha = np.mean(scores), alpha

    final_scaler = StandardScaler()
    X_tr_val_s = final_scaler.fit_transform(X_train_val)
    best_model = Ridge(alpha=best_alpha).fit(X_tr_val_s, y_train_val)
    preds = best_model.predict(final_scaler.transform(X_test))
    return best_model, preds, X_test.index
```

Module 3: Macro Trend and Regularized Momentum Filter

Harness File: `scripts/run_consolidated_showdown.py`

```
import numpy as np
import pandas as pd
from sklearn.linear_model import Ridge, Lasso
from sklearn.preprocessing import StandardScaler

def run_extension2(df_raw, test_index, tc=0.0010):
    df_m = df_raw.copy()
    df_m['Return'] = df_m['Price'].pct_change()
    df_m['Target_Next_Return'] = df_m['Return'].shift(-1)
    df_m['SMA_50'] = df_m['Price'].rolling(50).mean()
    df_m['SMA_200'] = df_m['Price'].rolling(200).mean()
    df_m['Dist_SMA50'] = (df_m['Price'] - df_m['SMA_50']) / df_m['SMA_50']
    df_m['Dist_SMA200'] = (df_m['Price'] - df_m['SMA_200']) / df_m['SMA_200']
    df_m['Mom_10d'] = df_m['Price'].pct_change(10)
    df_m['Mom_20d'] = df_m['Price'].pct_change(20)

    # Strategy 2a: Non-parametric 200-day trend rule
    pos_2a = np.where(df_m.loc[test_index, 'Price'] > df_m.loc[test_index, 'SMA_200'], 1.0, 0.0)

    # ML Momentum setup (Ridge and Lasso)
    m_feat = ['Dist_SMA50', 'Dist_SMA200', 'Mom_10d', 'Mom_20d']
    train_m = df_m.loc[df_m.index < test_index[0]].dropna()
    scaler = StandardScaler().fit(train_m[m_feat])
    y_tr = train_m['Target_Next_Return'].clip(-0.5, 0.5)

    ridge_m = Ridge(alpha=10.0).fit(scaler.transform(train_m[m_feat]), y_tr)
```

```
    lasso_m = Lasso(alpha=1e-5, random_state=42, max_iter=10000).fit(scaler.transform(train_m[m_feat]), y_tr)
    pos_2b = np.where(ridge_m.predict(scaler.transform(df_m.loc[test_index, m_feat])) > y_tr.mean(), 1.0, 0.0)
    pos_2c = np.where(lasso_m.predict(scaler.transform(df_m.loc[test_index, m_feat])) > y_tr.mean(), 1.0, 0.0)
    return pos_2a, pos_2b, pos_2c
```

Module 4: Dynamic Volatility Targeting Strategy

Source File: src/strategy_volatility.py

```
import numpy as np
import pandas as pd

def compute_volatility_features(df, vol_window=20, target_vol=0.40, epsilon=1e-6):
    df_out = df.copy()
    df_out['Daily_Return'] = df_out['Price'].pct_change()
    df_out['Target_Next_Return'] = df_out['Daily_Return'].shift(-1)
    rolling_std = df_out['Daily_Return'].rolling(window=vol_window, min_periods=vol_window).std()
    df_out['Rolling_Vol_20d'] = rolling_std * np.sqrt(252)
    safe_vol = np.maximum(df_out['Rolling_Vol_20d'], epsilon)
    df_out['Target_Weight'] = target_vol / safe_vol
    df_out['Position'] = np.clip(df_out['Target_Weight'], 0.0, 1.0)
    return df_out
```